# Benchmarking LLMs via Uncertainty Quantification

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# Objective

## Problem

Current LLM leaderboards (HuggingFace, OpenCompass . . . ) rank models by **accuracy only**.

Two models with *equal accuracy* may differ greatly in **uncertainty** – yet benchmarks treat them identically.

## Goal

Integrate **uncertainty quantification** into LLM benchmarking via **conformal prediction** – a statistically rigorous, distribution-free, model-agnostic method.

# Objective

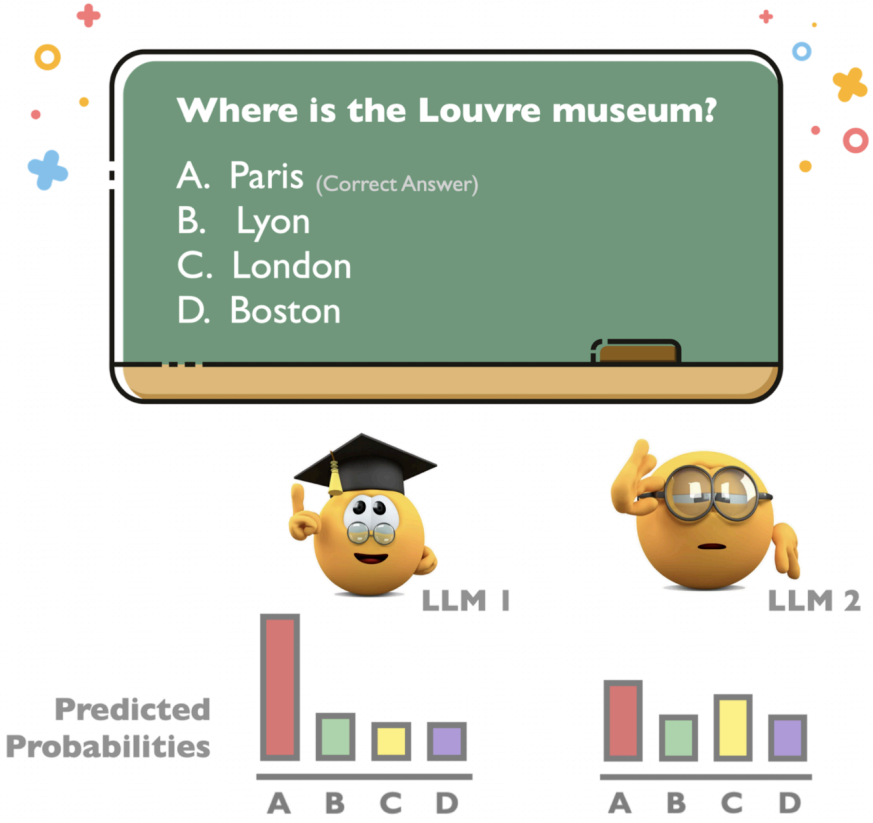


Figure: An illustration of two LLMs accurately predicting the true answer (with option A possessing the highest probability, but showing different levels of uncertainty. Note that when both LLMs predict a wrong answer, they may also display different levels of uncertainty.

# Methodology – Conformal Prediction

## Coverage Guarantee

For test instance $X_t$ with true label $Y_t$, CP outputs a **prediction set** $\mathcal{C}(X_t) \subseteq \mathcal{Y}$ such that:

$$P\big(Y_t \in \mathcal{C}(X_t)\big) \;\geq\; 1 - \alpha$$

$\alpha$ = user-specified error rate (e.g. $\alpha = 0.1 \;\Rightarrow\;$ **90 %** coverage guarantee).

## Uncertainty Measure

**Set Size (SS)** = average $|\mathcal{C}(X_t)|$ over all test instances.

Small SS $\Rightarrow$ **confident model** Large SS $\Rightarrow$ **uncertain model**

# Methodology – Score Functions

**LAC** – Least Ambiguous Classifier

$$s(X, Y) = 1 - f(X)_Y$$

Uses softmax score of the true label only.

$\Rightarrow$ tends to smaller sets.

**APS** – Adaptive Prediction Sets

$$s(X, Y) = \sum_{\substack{Y' \in \mathcal{Y}: \\ f(X)_{Y'} \geq f(X)_Y}} f(X)_{Y'}$$

Accumulates ranked softmax scores.

$\Rightarrow$ better on hard instances.

*Final metrics averaged over* **LAC** *and* **APS**.

# Algorithm – CP for LLMs

**Algorithm 1: Conformal Prediction for LLMs Input: $\mathcal{M}$, $\mathcal{D}$, $\alpha$ Output: Acc, SS, CR, UAcc**

**for each** prompt strategy $p \in \{\text{base, shared, task}\}$ **do** // *3 strategies*
  **for each** ICL setting $k \in \{\text{0-shot, few-shot, CoT}\}$ **do** // *3 ICL modes*
    $P(X) \leftarrow \mathrm{Softmax}(\mathrm{logits}_{\mathcal{M}}(X_{p,k}))$ **for all** $X \in \mathcal{D}$ // *query LLM*
    $\mathcal{D} \rightarrow \mathcal{D}_{\mathrm{cal}}$ (50%) $+ \mathcal{D}_{\mathrm{test}}$ (50%) // *random split*
    **for each** $s \in \{\text{LAC, APS}\}$ **do** // *2 score fns*
      $s_i \leftarrow s(X_c^{(i)}, Y_c^{(i)})$ **for all** $i \in \mathcal{D}_{\mathrm{cal}}$ // *nonconformity scores*
      $\hat{q} \leftarrow \mathrm{Quantile}\left(\{s_i\}, \frac{\lceil (n+1)(1-\alpha) \rceil}{n}\right)$ // *adaptive threshold*
      $\mathcal{C}(X_t) \leftarrow \{Y' \in \mathcal{Y} : s(X_t, Y') \leq \hat{q}\}$ **for all** $X_t \in \mathcal{D}_{\mathrm{test}}$ // *prediction sets*
      **if** $\mathcal{C}(X_t) = \emptyset$ **then keep** $\arg\max_{Y'} P(X_t)_{Y'}$
**return** avg over $p, k, s$: $\mathrm{Acc},\ \mathrm{SS},\ \mathrm{CR},\ \mathrm{UAcc} = \frac{\mathrm{Acc}}{\mathrm{SS}}\sqrt{|\mathcal{Y}|}$

# Methodology – Evaluation Metrics

| Metric | Formula | Meaning |
|---|---|---|
| Acc | $\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \mathbf{1}[Y_p = Y_t]$ | Prediction accuracy |
| SS | $\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \lvert\mathcal{C}(X_t)\rvert$ | Uncertainty (↑ more uncertain) |
| CR | $\frac{1}{\lvert\mathcal{D}_{\text{test}}\rvert}\sum \mathbf{1}[Y_t \in \mathcal{C}(X_t)]$ | Coverage ($\geq 1-\alpha$ guaranteed) |

**New: Uncertainty-aware Accuracy (UAcc)**

$$\mathrm{UAcc} = \frac{\mathrm{Acc}}{\mathrm{SS}}\sqrt{|\mathcal{Y}|}, \qquad |\mathcal{Y}| = 6$$

Rewards models that are both **accurate** and **certain**. Can **reverse rankings** vs. Acc alone.

# Results – All Base Models

Source: Table 1 – mean of LAC & APS, avg. over 5 tasks

| LLM | Acc (%) ↑ | SS ↓ | UAcc ↑ |
|---|---|---|---|
| Yi-34B | 81.46 | 1.93 | 121.09 |
| Qwen-72B | 78.05 | 2.06 | 109.96 |
| Qwen-14B | 74.00 | 2.17 | 102.28 |
| Llama-2-70B | 72.24 | 2.16 | 94.62 |
| DeepSeek-67B | 71.66 | 2.15 | 91.07 |
| Yi-6B | 68.97 | 2.36 | 83.53 |
| Mistral-7B | 64.14 | 2.43 | 70.84 |
| Llama-2-13B | 60.42 | 2.56 | 61.07 |
| Qwen-7B | 59.87 | 2.63 | 64.34 |
| InternLM-7B | 49.31 | 3.41 | 40.83 |
| Llama-2-7B | 46.53 | 3.09 | 38.71 |
| DeepSeek-7B | 45.87 | 3.13 | 36.80 |
| Qwen-1.8B | 42.34 | 3.38 | 32.95 |

# Results – Three Key Findings

**Finding 1:** Higher Accuracy $\neq$ Lower Uncertainty

InternLM-7B outperforms MPT-7B by **+19.34 %** Acc on DRS, yet shows **higher** SS (3.63 vs. 3.59). Rankings by Acc and SS diverge across *all* tasks.

**Finding 2:** Larger $\neq$ More Certain

Qwen 1.8B→14B: Acc and certainty both improve.
Qwen 14B→72B: Acc rises, but SS **increases** on RC & DRS – non-monotonic.

**Finding 3:** Instruction Finetuning Raises Uncertainty

Chat-V1 and Chat-V2 show **consistently higher SS** than base models across Llama-2, Yi, DeepSeek, and Falcon series.

# Results – CP vs. Perplexity (InternLM-7B)

**Table 2:** Coverage Rate (%) & Set Size – target CR $\geq$ 90 %

| | CR (%) | | SS | |
|---|---|---|---|---|
| **Task** | **CP** | **PPL** | **CP** | **PPL** |
| QA | **90.68** | 83.44 | 3.49 | 2.89 |
| RC | **93.28** | 95.48 | 2.19 | 2.39 |
| CI | **90.10** | 96.25 | 3.28 | 3.97 |
| DRS | **90.40** | 86.80 | 3.63 | 3.42 |
| DS | **90.34** | 87.13 | 4.47 | 4.33 |
| Avg | **90.96** | 89.82 | 3.41 | 3.40 |

PPL **fails** 90 % on QA, DRS, DS. CP **always meets** the coverage guarantee.

# Results – ECE Comparison (InternLM-7B)

**Table 3:** Expected Calibration Error (%) – lower is better

| Task | CP | Entropy | $P_{\max}$ |
|---|---|---|---|
| QA | 15.83 | 15.83 | 15.83 |
| RC | **1.33** | 1.32 | 1.41 |
| CI | **3.16** | 3.45 | 3.75 |
| DRS | **12.11** | 12.40 | 12.45 |
| DS | **9.30** | 9.30 | 9.62 |
| Avg | **8.35** | 8.46 | 8.61 |

CP achieves the **lowest average ECE** across all methods.

# Our Experiment – Qwen3.5-4B on MMLU

## Setup

- Model: `qwen3.5:4b` via Ollama (local inference)
- Dataset: MMLU ($n = 200$ test samples, 1-shot, base prompt)
- Logits via continuation logprob scoring (6 API calls/question)
- CP parameters: $\alpha = 0.1$, cal/test split 50 %

## Results

| Model | Acc (%) ↑ | SS ↓ | CR (%) |
|---|---|---|---|
| Qwen3.5-4B | 53.40 | 3.96 | **93.69** |

CR $\geq$ 90 % target **met**. SS = 3.96 reflects moderate uncertainty on MMLU.

# Conclusion & Future Work

## Conclusion

- Accuracy alone is **insufficient** for LLM evaluation.
- Conformal prediction offers a **rigorous, model-agnostic** uncertainty measure.
- New **UAcc** metric jointly captures accuracy and certainty.
- Three findings: Acc $\neq$ certainty; larger $\neq$ more certain; finetuning raises uncertainty.

## Future Work

- Extend CP to **free-form text generation** beyond MCQA.
- Apply to **multi-modal** and **closed-source** LLMs (GPT-4 etc.).
- Integrate uncertainty into **RAG** and **in-context learning** pipelines.